Phase I Design Document

Course: COMP 302 - Software Engineering
Group: International group

1. Vision

The vision for **Dungeon KUrawler** is to build a robust, extensible, object-oriented 2D game engine that supports a grid-based dungeon crawler. The primary game mode focuses on exploring randomly generated dungeons, puzzle-solving, combat, and resource management to find a specific hidden "Target Item." Crucially, the system will also feature a secondary "Build Mode" allowing players to design custom levels, emphasizing a modular architecture and high cohesion within the game's mechanics.

2. Functional Requirements

2.1. UML Use Case Diagram (Actors & Use Cases)

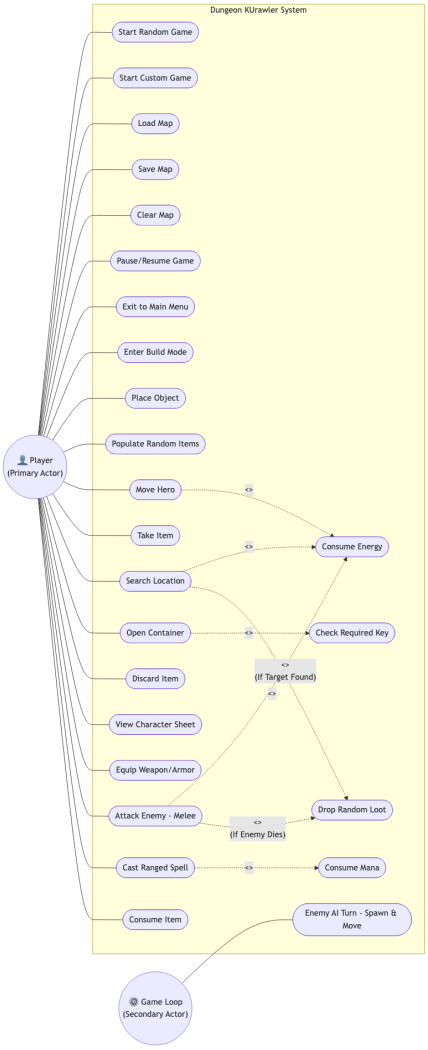

Figure 1: UML Use Case Diagram

Primary Actor: Player
Secondary Actor: Game Loop/Time (System)

- **Session & Level Management:** Start Random Game, Start Custom Game, Load Map, Save Map, Clear Map, Pause/Resume Game, Exit to Main Menu.
- **Editor/Build Mode:** Enter Build Mode, Place Object, Populate Random Items.
- **Hero Actions & Exploration:** Move Hero, Take Item, Search Location, Open Container, Discard Item.
- **Combat & Stats:** View Character Sheet, Equip Weapon, Attack Enemy (Melee), Cast Spell (Ranged), Consume Item.